
UNIT 1 INTRODUCTION TO SETS

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1.0 INTRODUCTION

Sometimes, we deal with some types of collections e.g.

- i) Collection of books in a university library..
- ii) Collection of natural numbers which are factors of, say, 20 or any other natural number.

A set is also a collection of objects, but it is a well-defined collection (we will learn more about this in Sec 1.2). Consider the collection of states in India. We know that there are 28 states in India, and this figure is exactly 28 (neither one more nor one less, as in January 2026). Also, if any number of people (with general knowledge of states in India) are asked to write the names of the states, the final list of everyone will contain the same 28 names (the order in which they write the names of the states does not matter). Such a well-defined collection is known as a set.

By **well defined** collection, we mean that given any object we must be able to know as to whether it belongs to the collection or does not belong to the collection.

In this unit, you will be introduced to the notations and terminology used for sets. The unit defines set, its various types, discusses hierarchy of sets, Venn diagrams, various operations on sets and finally, the unit is closed by giving an idea of some important and commonly used laws. Concepts related to sets are very elementary and are used directly or indirectly in many of the courses in this programme. So you need to understand the concepts discussed in this unit for further learning in this programme.

1.1 OBJECTIVE

After completing this unit, you should be able to:

- define a set,
- write a set in different forms,
- explain the types of sets,
- draw Venn diagrams,
- apply different operations on sets,
- get the idea of super sets and subsets,
- get an idea of some important laws related to sets, like commutative, associative, De-Morgan's laws, etc., and
- apply set theory concepts for problem-solving.

1.2 SETS

Conventionally, the concept of *number* and geometric concepts like '*straight line*', '*triangle*' and '*square*'. have been the most fundamental, and the very first of the concepts discussed in introductory mathematics. Since almost the middle of the previous century, the concept of '**set**' is considered as the most fundamental concept of mathematics and all other concepts, including even those of counting numbers and geometrical shapes, are discussed in terms of **set**.

1.2.1 Concept of Set

Set is a basic concept like 'point', & 'straight line' in geometry and like 'number' in arithmetic, and hence, should be left undefined. However, in view of the fact, that 'set' is now-a-days considered as one of the most fundamental concepts in mathematics; to make a beginning in *new* mathematics, we just give a sort of working definition, and define some elementary operations on sets to be frequently used in the subsequent discussion.

Roughly, a set may be considered a collection of things or entities, such as names of animals, names of books, names of numbers, names of geometrical shapes, or any mixture of these. Any entity in a set is called a member, or an element, of that set. Further, in a set, **no member is written more than once**. Also, the entities must be **distinctly identifiable**. For example, as the drops of water in a glass cannot be distinctly identified, hence, drops of water in a glass **do not** form a set.

More formally, we take a **set as a well-defined collection of objects**. *What is meant by 'well-defined' needs to be understood clearly. A collection is said to be well-defined, if:*

- (i) entities included in it are distinctly identifiable, and,
- (ii) if we pick up any distinctly identifiable entity in the world, then we should be able to tell whether the picked-up entity is in (i.e. belongs to) the collection or not.

For example, let us consider the collection of **all** students of IGNOU who enrolled in the 2026 July session for the B.A. program. This is a well-defined collection, hence a set. But if we say the collection of **some of** those students of

IGNOU who enrolled in the 2026 July session for the B.A. program, then **it is NOT a set**. Also, if we say 'collection of all drops of water in this glass', then **it is NOT a set**, because, as mentioned earlier, a 'drop of water' is not distinctly identifiable. Thus, **not any collection is a set, only a well-defined collection is a set**.

A set is generally denoted by capital letters such as A, B, C, X, Y, Z, etc. and the objects which belong to the set are known as elements or members of the set and are generally denoted by small letters a, b, c, x, y, z, etc.

If 'a' is an element of a set A then we write $a \in A$ (read it as 'a' belongs to A). If 'a' is not an element of A then we write $a \notin A$ (read as 'a' does not belong to A).

Following example illustrates the term “well defined collection” or “set”.

Example 1: Consider the following collections and state reasons whether they form set or not.

- (i) Collection of good cricketers in India.
- (ii) Collection of honest students in a particular university in India.
- (iii) Collection of natural numbers which are less than 5.
- (iv) Collection of rich persons in India.
- (v) Collection of letters of the word “ASSIGNMENT”

Solution:

- (i) This collection does not form a set, because a given player may be good according to some person, but the same player may not be good according to some other person.
- (ii) This collection does not form a set, because a student may be honest according to some person, but the same student may not be honest according to some other person.
- (iii) Yes, this collection forms a set, and its elements are 1, 2, 3, and 4.
- (iv) Richness is not a well-defined property, because, according to one person, a person may be rich while he/she may not be rich in view of some other person. So this collection does not form a set.
- (v) It is a set, and its elements are A, S, I, G, N, M, E, T.

Notations

Like most of the mathematical concepts, sets have special notation. For illustration, the **notation** $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ is used to denote the set of all decimal digits, viz., 0, 1, 2, 3, 4, 5, 6, 7, 8, 9.

A set is generally represented by two methods: the **roster method** and **set-builder method**.

The notation for a set has at the leftmost position a left brace, denoted by '{'. Followed by one of the elements, say 0, then a comma and a blank-space, next another element say 1, followed by a comma and blank-space, and so on, till the last element is written, which is followed by the right-brace, denoted by '}'. Generally, a set is given a name, e.g., in this case the name of the set may be

Num-Digits, then we write $\text{Num-Digits} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$. Any distinct name –the name which has not been used earlier for naming any other set –may be assigned. However, in general, the name assigned is mnemonic

“a well-defined collection of distinct objects is called a set”.

A set remains the same if some or all of its elements are repeated or rearranged.

For example, if a set contains the elements 0, 1, – 1 and another set contains the elements – 1, – 1, 1, 0, 0, 0, 1, 0, 1, then these two sets are nothing but represent the same set having three elements 0, 1, – 1.

(i.e., it should help remember its contents or members).

Member/element of a set is what is included, except commas, within braces. For example, 0 is a member, or element, of Num-Digits, 5 is a member of Num-Digits etc. Instead, we may also say 0 **belongs to** Num-Digits, 5 *belongs to* Num-Digits etc. However, we say 55 *does not belong to* Num-Digits.

Notation for member is the symbol $\in$, called *epsilon*. It is used to denote membership of a set. For the above example of Num-Digits, we will write $0 \in \text{Num-Digits}$, $9 \in \text{Num-Digits}$ etc. Similarly, if we have a set called Vowels, with $\text{Vowels} = \{a, u, e, o, i\}$, then $e \in \text{Vowels}$, $u \in \text{Vowels}$ etc. *Whenever x is a member of the set X , or equivalently, when $x \in X$, then we also say ' x belongs to X '. The notation for 'is not a member of' or for 'does not*

belong to' is $\notin$. For example, for the definition of Num-Digits given above, $55 \notin \text{Num-digits}$.

Extension of Set Notation: The problem with the set notation mentioned above arises when the set has either a very large or an infinite number of elements. Because then it is quite difficult, if not impossible, to specify all the elements required to use the notation. In such cases, if there is some pattern among the elements, then only some initial members are mentioned, followed by **triple dots(...)**. For example, the set of numbers, each of which is less than 10,001 and is also square of some integer, may be denoted by $\{0, 1, 4, 9, \dots, 100, 121, \dots, 400, \dots, 10000\}$, or even more briefly by $\{0, 1, 4, 9, \dots, 10000\}$ etc. It may be named as **Sq_int_less_10001**.

1.2.2 Methods of Representing a Set

A set is generally represented by two methods as given below.

A. Roster Method

Dictionary meaning of 'Roster' is 'a list showing persons who perform their duties in turn'. As its meaning suggests, in this method each and every element is listed and put, separating by commas, in curly brackets. This method is also known as **Tabular Form** or **Listing Method**.

For example,

- (i) If A is the set of vowels of English alphabets, then $A = \{a, e, i, o, u\}$
- (ii) If N is the set of natural numbers, then $N = \{1, 2, 3, 4, 5, \dots\}$
- (iii) If W is the set of whole numbers, then $W = \{0, 1, 2, 3, 4, 5, \dots\}$
- (iv) If Z is the set of integers, then $Z = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$
- (v) If E is the set of even natural numbers, then $E = \{2, 4, 6, 8, 10, 12, \dots\}$
- (vi) If O is the set of odd natural numbers, then $O = \{1, 3, 5, \dots\}$
- (vii) If P is the set of prime numbers, then
 $P = \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, \dots\}$

Remark 1:

Throughout the course, we use N, W and Z for the sets of natural numbers, whole numbers, and integers, respectively.

B. Set-Builder Method

In this method, we consider one or more properties that are exclusive to the elements of a set, so that no other elements can be members of the set. This method is also known as the **Property Method** or **Rule Method**.

For example,

- (i) Let $A = \{x : x \text{ is a vowel of English alphabet}\}$, then elements of A are a, e, i, o, u and having exclusive property of being a vowel no other alphabet can be considered as an element of set A .
- (ii) Let $A = \{x : x \text{ is a natural number and } x \text{ is a multiple of } 3\}$, then elements of A are 3, 6, 9, 12, ... which have the exclusive property of being multiple of 3 and no other element can be consider as an element of A .
- (iii) Let $A = \{x : x \text{ is a factor of } 10 \text{ and } x > 0\}$, then elements of A are 1, 2, 5, 10 and having exclusive property of being a factor of 10 and no other element can be consider as an element of A .
- (iv) If $\mathbb{Q}$ is the set of rational numbers then $\mathbb{Q} = \left\{x : x = \frac{p}{q}, p, q \in \mathbb{Z}, q \neq 0\right\}$

Rational number: A number which can be expressed in the form $\frac{p}{q}$ such that $p, q \in \mathbb{Z}, q \neq 0$ is known as rational number. For example, $\frac{2}{3}, \frac{-4}{13}, \text{etc.}$

Remark 2:

- (i) In terminology of set the symbol “:” used in each part above is read as “such that”.
- (ii) Throughout the course, the sets of rational numbers, irrational numbers, and real numbers will be denoted by $\mathbb{Q}$, $\mathbb{I}$, and $\mathbb{R}$, respectively.

Note: Advantage of the second method, i.e. Set-Builder method lies in the fact that sometimes (or in some situations) we cannot list the elements of the set or even if we can list them, it may not be practical or feasible to do so.

For example, consider the set $\{x : x \text{ is a person who was born in 2010 in India}\}$. Obviously, you will be more comfortable with the property method in this example.

Consider another set $\{x : x \text{ is a real number and } 1 < x < 4\}$. This set cannot be described by the listing method, because the number of elements(Cardinality) in this set is uncountable.

Thus, the above two examples show that in some situations, either it is too difficult to describe the set by the listing method or it is impossible to describe it.

No doubt, there are some examples in which the listing method has its advantages .

For example, consider the set $\{28, \text{Bihar, India}\}$.

It is a set having three elements 28, Bihar and India.

Let us now consider some examples to clarify the ideas of the two methods discussed above.

Example 2: Write the following sets by the roster method:

- (i) $A = \{x : x \text{ is a letter of the word “FUNCTION”}\}$
- (ii) $B = \{x : 2x + 5 < 17, x \in \mathbb{N}\}$
- (iii) $C = \{x : x^2 - x - 12 = 0, x \in \mathbb{N}\}$
- (iv) $D = \{x : x^2 - 4x - 21 = 0, x^2 - 49 = 0, x \in \mathbb{N}\}$

Real number: A number which is either rational or irrational is called real number.

Solution:

(i) $A = \{F, U, N, C, T, I, O\}$ [Repeated elements are written once only]

(ii) $B = \{x : 2x < 17 - 5, x \in \mathbb{N}\} = \{x : 2x < 12, x \in \mathbb{N}\}$
 $= \{x : x < 6, x \in \mathbb{N}\} = \{1, 2, 3, 4, 5\}$

(iii) $C = \{x : x^2 - x - 12 = 0, x \in \mathbb{N}\} = \{x : x^2 - 4x + 3x - 12 = 0, x \in \mathbb{N}\}$
 $= \{x : x(x - 4) + 3(x - 4) = 0, x \in \mathbb{N}\} = \{x : (x - 4)(x + 3) = 0, x \in \mathbb{N}\}$
 $= \{x : x = 4, -3, x \in \mathbb{N}\} = \{4\} \quad [\because -3 \notin \mathbb{N}]$

(iv) $D = \{x : x^2 - 4x - 21 = 0, x^2 - 49 = 0, x \in \mathbb{N}\}$
 $= \{x : x^2 - 7x + 3x - 21 = 0, x^2 - 7^2 = 0, x \in \mathbb{N}\}$
 $= \{x : x(x - 7) + 3(x - 7) = 0, (x - 7)(x + 7) = 0, x \in \mathbb{N}\}$
 $= \{x : (x - 7)(x + 3) = 0, x = 7, -7, x \in \mathbb{N}\}$
 $= \{x : x = 7, -3, x = 7, -7, x \in \mathbb{N}\}$
 $= \{7\}$

($\because$ here are two properties one says $x = 7, -3$ and second says $x = 7, -7$, but in case we have more than one property, we take common element(s) between them, and in our case, it is 7.)

Example 3: Express the following sets in the set-builder form:

(i) $A = \{3, 6, 9, 12, 15, 18, \dots\}$

(ii) $B = \{1, 2, 3, 5, 6, 10, 15, 30\}$

(iii) $C = \{5, 25, 125, 625, \dots\}$

(iv) $D = \{1, \frac{1}{3}, \frac{1}{9}, \frac{1}{27}, \frac{1}{81}, \frac{1}{243}\}$

(v) $E = \{1, 4, 9, 16, 25, 36, 49, 64, 81, 100\}$

Solution:

(i) Here we see that elements in this set are multiple of 3 so in set-builder form it can be written as $A = \{x : x \text{ is a multiple of } 3, x \in \mathbb{N}\}$.

Similarly, by observing the pattern obeyed by the elements of other parts we can write them as given below.

(ii) $B = \{x : x \text{ is a factor of } 30, x \in \mathbb{N}\}$

(iii) $C = \{x : x = 5^n, n \in \mathbb{N}\}$

(iv) $D = \{x : x = \frac{1}{3^n}, 0 \leq n \leq 5, n \in \mathbb{W}\}$

(v) $E = \{x : x = n^2, 1 \leq n \leq 10, n \in \mathbb{N}\}$

1.2.3 Relationships between Sets

For any two sets X and Y, the set relationships that we discuss are:

- (i) equality (=) of X and Y
- (ii) X is a subset/ superset ($\subseteq$) of Y
- (iii) X is a proper subset/superset($\subset$) of Y, and
- (iv) is not a subset of ($\not\subset$).

1.2.3.1 Notations, Definitions and Examples

- i) Two sets X and Y are said/defined as **equal** if every element of X is an element of Y , *and* every element of Y is an element of X .

The notation $X=Y$ is used for 'equality of sets X and Y '

Example 4: Let $X = \{1, 2, 3, 4, 5\}$, and $Y = \{x: x \text{ is a natural number such that } 1 \leq x \leq 5\}$, and $W = \{3, 5, 2, 4, 1\}$. Then $X = Y = W$.

- ii) X is said to be a **subset** of Y if every element of X is an element of Y .
Instead of saying ' X is a subset of Y ' we may equivalently say ' Y is a **superset** of X ', or ' X is contained in Y ' or ' Y contains X '.

The notation $X \subseteq Y$ is used for ' X is a subset of Y '.

Example 5:

(i) Let $X = \{a, e, i\}$, and $Y =$ set of all vowels in English language. Then $X \subseteq Y$.

(ii) Every set is its own subset, i.e. $X \subseteq X$ (can be easily verified by applying the definition).

iii) X is said to be a **proper subset** of Y if (i) every element of X is an element of Y , *and also* (ii) there is at least one element of Y which does not belong to X . Then, it can also be said that Y is a proper superset of X .

Definition: A set A is said to be a **proper subset** of a set B if A is a subset of B and A and B are not equal. We represent this by $A \subset B$.

For example, if $A = \{4, 5, 6\}$ and $B = \{4, 5, 7, 8, 6\}$, then $A \subset B$; and if $A = \{\text{Java, C, C++, Cobol}\}$ and $B = \{\text{Java, C++}\}$, then $A \supset B$.

Note: A set can have many subsets and many supersets.

For example, $A = \{1, 2, 3, 4, 5\}$, $B = \{2, 3, 4, 5, 6, 7\}$, and $C = \{2, 3\}$, then for C , A and B can be used as supersets.

Similarly, if $X = \{\text{Ram, Rani, Sita, Gita}\}$, $Y = \{\text{Rani}\}$, and $Z = \{\text{Sita}\}$, then Y and Z both are subsets of X .

The notation $X \subset Y$ is used for ' X is a proper subset of Y ' and also for ' Y is a proper superset of X '.

Example 6: Let $X =$ set of even decimal digits $= \{0, 2, 4, 6, 8\}$ and let $Y =$ set of all decimal digits $= \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$. Then $X \subset Y$, as every even decimal digit is also a digit. But $1 \in Y$, and $1 \notin X$.

The notation $X \not\subset Y$ denotes the fact that X is **not a subset** of Y . For showing $X \not\subset Y$, we need to show *only* that there is at least *one* element of X which does not belong to Y . For example, Let $X = \{1, 5, 10, 15\}$, and $Y = \{1, 12, 14, 15\}$. To show $X \not\subset Y$, we need *only* to show that $5 \in X$, and $5 \notin Y$. We need not go further to say that $10 \in X$, and $10 \notin Y$. Also, $X \not\subset X$, as condition (ii) for being a proper subset, is not satisfied.

1.2.3.2 Establishing Set Relationships

Here, through a simple example, we demonstrate how to establish each of these set relationships. Later, we will consider some more complex examples of establishing these relationships, particularly regarding the equality (=).

Let us consider sets

$A = \{x: x^2+3x+2=0\}$, the set of solutions of the equation $x^2+3x+2=0$.

$B = \{-1, -2\}$, $C = \{-2\}$, $E = \{0, -1, -2\}$

- i) Easiest case is that of '**is not a subset of**': In our case, $E \not\subseteq C$, as $0 \in E$, but $0 \notin C$. Similarly, $E \not\subseteq A$, as $0 \in E$, but 0 is not a root of $x^2+3x+2=0$.
- ii) Next, in order to show $B \subseteq E$, we need to consider, one by one, each element of B , and show that the considered element belongs to E also. In complex cases, we do not consider each element one by one, but take an arbitrary element.
- iii) Further, in order to show that B is a **proper subset** of E (i.e. $B \subset E$), in addition to what we have done in (ii) above, we need to show the existence of an element of E , which does not belong to B . In our case $0 \in E$, but $0 \notin B$.
- iv) To show equality (=) of two sets, we need to repeat case (ii) two times, reversing the roles of the sets under consideration.
In our case, $A=B$, because -1 , which belongs to B , is a root of $x^2+3x+2=0$, hence, belongs to A . Similarly, -2 , which belongs to B , is a root of $x^2+3x+2=0$, hence, belongs to A . Conversely, as $x^2+3x+2=0$ is a quadratic equation, it has exactly two roots, viz. $-1, -2$.

1.2.4 Cardinality of a Set

The number of elements in a finite set say A is called its Cardinality or cardinal number of a finite set and is denoted by $n(A)$.

For example,

- (i) If $A = \{x, y, z\}$, then $n(A) = 3$, i.e. cardinality of A is 3.
- (ii) If $B = \{1, 4, 2, 3, 9, 15\}$, then $n(B) = 6$, i.e. cardinality of B is 6.

1.2.5 Subsets

Before we relook at the subset, let us define the empty set.

The **empty set** (also known as the null set) is a unique concept in mathematics and set theory. It represents a set that contains **no elements**. It is denoted by $\emptyset$ or $\{\}$: A pair of curly braces with nothing between them.

Definition: A set A is said to be a **subset** of a set B if each element of A is also an element of B . In this case, B is called a **superset** of A . If A is a subset of B , we represent this by $A \subseteq B$.

As a statement in logic, we represent this situation as,

$$A \subseteq B \iff [x \in A \Rightarrow x \in B]$$

' B contains ' A ' or ' B ' is a superset of A ' is represented by $B \supseteq A$.

If A is not a subset of B, we write $A \not\subseteq B$.

For example,

- i) if $A = \{4, 5, 6\}$ and $B = \{4, 5, 7, 8, 6\}$, then $A \subseteq B$. But if $C = \{3, 4\}$ then $C \not\subseteq B$.
- ii) If set A is set of all the cricketers of Indian Cricket Team and set B is all the bowlers of India cricket Team then $B \subseteq A$.

Remarks 2:

i) There is **no complete standardization of the notation of subset**. Some people use $\subset$ to denote a subset, which may or may not be a proper subset. Then, they use the symbol $\subsetneq$ for proper subset. We could have used these symbols. But we would continue with symbols ' $\subseteq$ ' and ' $\subsetneq$ ' denoting respectively 'subset' and 'proper subset'.

Next, we state, without proofs, some important results about the relation of subset

- $A \subseteq A$ for every set A. Every set is a subset of itself.
- If $A \subseteq B$ and $B \subseteq C$ then $A \subseteq C$. We say that $\subseteq$ is a *transitive relation* between sets.
- If $A \subseteq B$ and $B \subseteq A$ then $A = B$.

1.2.6 Power Set of a Set

Definitions: The power set of a set A is the set of all the subsets of A, and is denoted by $P(A)$. Sometimes power set of a given set A is denoted by 2^A , because that is the Cardinality of the power set A.

Mathematically, $P(A) = \{x : x \subseteq A\}$.

Note that:

- If $|A| = n$, then $P(A) = 2^n$. For example, the set $\{1, 2, 3\}$ has 3 elements, and, the power set of A, viz. $p(A) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$ has $2^3 = 8$ elements.
- $\emptyset \in P(A)$ and $A \in P(A)$ for all sets A.

For example, if $A = \{1\}$, then $P(A) = \{\emptyset, \{1\}\}$ and if $A = \{1, 2\}$, then

$P(A) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$

Similarly, if $A = \{1, 2, 3\}$, then $P(A) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$.

Definition: Any set which is a superset of all the set under consideration is known as the **universal set**. This is usually denoted by Ω , S or U.

For example, if $A = \{1, 2, 3\}$, $B = \{3, 4, 6, 9\}$ and $C = \{0, 1\}$, then we can take

$U = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ or $U = \mathbb{N}$, or $U = \mathbb{Z}$ as the universal set.

Note that the universal set can be chosen arbitrarily for a given problem. But once chosen, it is fixed for the discussion of that problem.

Theorem 1: If A is a set with n elements, then $|P(A)| = 2^n$.

Proof: We shall prove this by mathematical induction.

For this, we first check if it is true for $n=1$. Then assuming that it is true for $n=m$, we prove it for the case $n = m + 1$. It will, then, follow that the result will be true $\forall n \in \mathbb{N}$.

Step I: If $A = 1$, then $P(A) = 2 = 2^1$.

Step II: Assume that the theorem holds for all sets A of cardinality k , i.e. if $|A| = k$, then A has 2^k subsets.

Step III: Now consider any set $A = \{x_1, x_2, x_3, \dots, x_k, x_{k+1}\}$, with $k+1$ elements. Consider its subset $B = \{x_1, x_2, x_3, \dots, x_k\}$. Now B has 2^k subsets, each being a subset of A . Now, take any such subset $\{x_{i1}, x_{i2}, \dots, x_{ir}\}$ of B . Then $\{x_{i1}, x_{i2}, \dots, x_{k+1}\}$ is a subset of A that is not a subset of B . So, for each of the 2^k subsets of B , we attach x_{k+1} to it to get 2^k more subsets of A .

You can see that this covers all the subsets of A .

So the number of subsets of $A = 2^k + 2^k = 2 \cdot 2^k = 2^{k+1}$.

Hence the theorem.

Let us first consider some examples:

- (i) If $A = \{ \}$, then $\emptyset$ is only subset of A .
That is, there is only 1 ($=2^0$) subset of A .
- (ii) If $A = \{a\}$, then possible subsets of A are $\emptyset, \{a\}$.
That is, there are only 2 ($=2^1$) subsets of A .
- (iii) If $A = \{a, b\}$, then possible subsets of A are $\emptyset, \{a\}, \{b\}, \{a, b\}$.
That is, there are only 4 ($=2^2$) subsets of A .
- (iv) If $A = \{a, b, c\}$, then possible subsets of A are $\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}$.
That is, there are only 8 ($=2^3$) subsets of A .

For example, in the above discussed cases (i) to (iv), the power set of A is given by

- (i) $P(A) = \{ \emptyset \}$
- (ii) $P(A) = \{ \emptyset, \{a\} \}$
- (iii) $P(A) = \{ \emptyset, \{a\}, \{b\}, \{a, b\} \}$
- (iv) $P(A) = \{ \emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\} \}$

Check Your Progress -1

- 1) Give reasons whether the following collections are sets or not.
 - (i) Collection of intelligent students in a particular school.
 - (ii) Collection of good hockey players in India.
 - (iii) Collection of good actors in India.

(iv) Collection of vowels in the word "INDIA".

.....

2) Describe the following sets by roster method:

(i) $A = \{x : x = 2n + 3, n \in \mathbb{W}\}$

(ii) $B = \{x : x = 7^n, 0 \leq n \leq 3, n \in \mathbb{N}\}$

(iii) $C = \{x : x \in \mathbb{N} \text{ and } x \in \mathbb{W}\}$

(iv) $D = \{x : x \in \mathbb{W} \text{ and } x \in \mathbb{Q}\}$

.....

3) Express the following sets in set-builder form:

(i) $\{5, 10, 15, 20, \dots\}$ (ii) $\left\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\right\}$ (iii) $\{2, 4, 6, 8, 10, \dots\}$

.....

4) For the pair of sets X and Y, state whether $X = Y$ or not, along with explanation for your response.

i) $X = \{2, 4, 6, 8, 10\}$, $Y = \{x : x \text{ is positive even integer and } x \leq 10\}$

ii) $X = \{x : x \text{ is a multiple of } 10\}$, $Y = \{10, 15, 20, 25, 30, \dots\}$

iii) $X = \{x : x \text{ is a letter in the word FOLLOW}\}$, $Y = \{y : y \text{ is a letter in the word WOLF}\}$

.....

5) For $X = \{1, \{2, 3\}, 3, 4\}$, which ones of the following is/are true statement(s)?

(i) $2 \in X$, (ii) $\emptyset \in X$, (iii) $\emptyset \notin X$, (iv) $\emptyset \subset X$, (v) $\{2, 3\} \in X$, (vi) $\{2, 3\} \subset X$, (vii) $\{\{2, 3\}\} \subset X$

.....

6) Give two proper subsets and two supersets of the set of vowels of the English alphabet.

.....

7) Find the power set of the set $A = \{a, e, i, o, u\}$.

.....

8) For which set A, is $P(A) = 1$?

.....

9) If $A \subseteq B$, is $P(A) \subseteq P(B)$? Why?

.....

10) $P(A) = P(B) \implies A = B$. True or false? Why?

.....

1.3 TYPES OF SETS

We have seen that set is a well-defined collection of distinct objects. Also repetition of elements in a set is not allowed. So once a set is defined, automatically number of elements contained by it has also become fixed. In this section, we shall discuss the different names given to a set on the bases of the number of elements contained by the set. Equivalent and equal sets are also defined in this section.

Null Set or Empty Set

Consider a collection of those sons having their ages more than their respective fathers. Of course we will find no such son in this world. This type of collection is nothing but simply known as null set or **empty set** or **void set** in the terminology of sets.

It may appear strange, at least when heard for the first time, that a set may have no member at all, or the number of elements in a set may be zero. According to our definition of a set, the following is a set:

$S = \{x: x \text{ is an even integer, and } x^2 = 9\}$.

The collection S is well-defined, and hence, is a set. But there is no member which satisfies the conditions of its membership. Therefore, the number of members of S is zero. There are many combinations of conditions, which may make a set to have no member, Each of such sets is called a Null set, a Void set or an empty set. **A null set is denoted by the standard notation $\emptyset$ or by $\{\}$.**

Remarks 3: $\emptyset$ is a subset of every set X , i. e. $\emptyset \subseteq X$, for every set X .

The fact is easily established through contradiction, a common proof technique in Mathematics: Assume what is to be proved as wrong, and then reach a fact which conflicts with some earlier established fact. So, let $\emptyset \subseteq X$ be false. Then there must be an $x \in \emptyset$, and $x \notin X$. But $x \in \emptyset$ is not possible by definition of $\emptyset$. Hence, the assumption that ' $\emptyset \subseteq X$ be false' is wrong. Hence the proof.

Let us now formally define null set.

A set is said to be null (or empty or void) if it has no element in it. Null set is denoted by $\emptyset$ or $\{\}$.

For example, $A = \{x : x \text{ is a natural number, } 1 < x < 2\}$ is a null set as there is no natural number between 1 and 2.

Proper subsets of a set X: As $\emptyset$ and a set X itself are subsets of the set X. These two subsets are not considered as proper subsets. All other subsets are considered as proper subsets.

Singleton Set

Consider the collection of mothers of a baby. Obviously, a baby has only one mother. This type of collection having a single element is known as singleton set in the terminology of sets. Thus, a singleton set is defined as follow.

A set is said to be singleton set if it contains only one element.

For example, $A = \{x : x \text{ is an even prime number}\}$ is a singleton set as there is only one even prime number, i.e. 2.

Finite Set

A set is said to be finite set if either it is an empty set or it has a finite number of elements.

For example,

- (i) $A = \{2, 5, 7, 15\}$ is a finite set because it contains 4 elements, i.e. finite number of elements.
- (ii) $B = \{1, 2, 3, 4, 5, \dots\}$ is not a finite set as the number of elements in it are infinitely many.
- (iii) $C = \{x : x + 1 = 0, x \in \mathbb{N}\} = \{x : x = -1, x \in \mathbb{N}\} = \{\}$ is an empty set. So, it is a finite set.

Infinite Set

A set is said to be infinite if it is not finite.

For example, $A = \{1, 4, 9, 16, 25, 36, \dots\}$ is an infinite set.

Equivalent Sets

Two finite sets A and B (say) are said to be equivalent if number of elements in both the sets are equal in numbers, i.e. $n(A) = n(B)$ and we denote it by $A \sim B$ (read as A is equivalent to B).

For example, if $A = \{a, b, c, d\}$ and $B = \{2, 3, 5, 7\}$, then
 $A \sim B$ [$\because n(A) = n(B) = 4$]

Equal Sets

Two sets A and B are said to be equal if every element of A is in B and every element of B is in A and is written as $A = B$.

If two sets A and B are not equal then we write $A \neq B$.

Definition: Two sets A and B are equal if every element of A belongs to B and every element of B belongs to A. We represent this by $A=B$.

For example,

- i) if $A = \{a, b, c, d\}$ and $B = \{c, b, d, a\}$, then $A = B$ as order of elements does not matter.
- ii) if $A = \{1, 2, 3\}$, $B = \{2, 3, 1\}$, then $A \subset B$ and $B \subset A$, so that $A = B$.
- iii) if $A = \{1, 3, 5, 7, 9\}$, $B = \{2, 3, 5\}$ then $A \neq B$.

Example 7: Give reasons whether the following statements are true or false:

- (i) If $A = \{x : x \text{ is a vowel of English alphabet}\}$ and $B = \{x : x \text{ is a natural number, } 7 < x < 13\}$, then $A = B$
- (ii) If $A = \{x : x^2 = 9, x \in \mathbb{Z}\}$ and $B = \{3, -3\}$, then $A = B$
- (iii) If $A = \{x, y, z, w\}$ and $B = \{d, e, 7, 9\}$, then $A \sim B$
- (iv) If $A = \{x, x, y, z\}$ and $B = \{x, y, z, w\}$, then $A \sim B$

Solution:

(i) Here, $A = \{a, e, i, o, u\}$ and $B = \{8, 9, 10, 11, 12\}$.
Clearly, $A \neq B$ (as their elements are not same). Hence, the given statement is false.

(ii) Here, $A = \{3, -3\}$

$$\left[\begin{array}{l} \because \text{we know that if } x^2 = a \text{ then } x = \pm\sqrt{a} \\ \therefore x^2 = 9 \Rightarrow x = \pm\sqrt{9} = \pm 3 \end{array} \right]$$

and

$B = \{3, -3\}$, Clearly, $A = B$

$$\left[\begin{array}{l} \because \text{all the elements of } A \text{ are in } B \text{ and} \\ \text{all the element of } B \text{ are in } A. \end{array} \right]$$

$\therefore$ the statement is true.

(iii) Here, $n(A) = 4$, $n(B) = 4 \Rightarrow A \sim B$, therefore, it is a true statement.

(iv) Here, $A = \{x, x, y, z\} = \{x, y, z\}$ [As repetition in a set is not allowed]
and $B = \{x, y, z, w\}$, so $n(A) = 3$, $n(B) = 4$.
 $\Rightarrow A \sim B$, is false because $n(A) \neq n(B)$.

1.4 HIERARCHY OF SETS

For given any two real numbers a and b , you know that either $a = b$ or $a < b$ or $a > b$. This section will focus on how this type of association is setup in case of sets. Actually, here, we consider the sets contained in some other sets and define them with their appropriate designations.

Subset

Suppose A be the set of all working ladies and B be the set of all ladies then obviously all working ladies are ladies first. That is all the members of the set A are members of the set B . If it is so, then in the terminology of the sets A is known as subset of B .

Now, let us formally define the term subset.

Let A and B be two sets. Then A is said to be subset of B (or B is **super set** of A) if every element of A belongs to B and is denoted by $A \subseteq B$.

“ $A \subseteq B$ ” read as A is contained in B or A is a subset of B .

If we write it as “ $B \supseteq A$ ” then we read it as B contains A and we call B is a super set of A .

Remark 4: From above definition of subset, we see that a set A (say) will not be subset of another set B (say) if there is at least one element in A which is not in B . And it is denoted by $A \not\subseteq B$ (read as A is not a subset of B or A is not contained in B).

For example,

- (i) If $N = \{1, 2, 3, 4, \dots\}$, $W = \{0, 1, 2, 3, 4, 5, \dots\}$,
 $Z = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$, $\mathbb{Q} = \{\frac{p}{q} : p, q \in Z, q \neq 0\}$,
 then $N \subseteq W$, $W \subseteq Z$, $Z \subseteq \mathbb{Q}$, i.e. $N \subseteq W \subseteq Z \subseteq \mathbb{Q}$.
- (ii) If $A = \{1, 2, 4\}$, $B = \{1, 2, 4, 7, 9\}$, then $A \subseteq B$
- (iii) If $A = \{a, b, c\}$, $B = \{b, c, d, e\}$ then $A \not\subseteq B$ [$\because a \in A$ but $a \notin B$]

Proper Subset

Let A and B be two sets. Then A is said to be proper subset of B if all the elements of A are in B and B has at least one element other than elements of A and is denoted by $A \subset B$.

For example, if $A = \{1, 2, 3\}$ and $B = \{1, 2, 3, 4, 5\}$,

then $A \subset B$. [$\because$ all the elements of A are in B and B has two extra elements, i.e. 4 and 5.]

Remark 5:

- (i) Empty set is subset of every set, i.e. $\emptyset \subseteq A$ for any set A.
- (ii) Every set is a subset of itself, i.e. $A \subseteq A$ for every set A.

Universal Set

Universal Set: For discussion problems relating to educating people in India, we need not take into consideration all human beings, we need to consider set of only Indians. While using mathematics for solving problems (like educating Indians) from some domain of human experience, some sets of entities from the domain are considered as and when required. For facilitating arguments, a sufficiently large set of entities from the problem domain- which includes all possible entities that may be required at any time during the discussion or solution-process of solving the problems of the type under consideration- is taken. For example, in our present case, the set of all Indians. **Such a set is called the Universal set and is generally denoted by U.** Thus, by definition, every set X under consideration will be a subset of the Universal set U. i. e. $X \subseteq U$ for all sets x under consideration during solving a particular (type of) problem.

Let us take IGNOU, there are 21 schools such as school of humanities (SOH), school of social sciences (SOSS), school of sciences (SOS), etc. (source IGNOU dairy 2011). If U is the set of all faculties of IGNOU and $A_1, A_2, A_3, \dots, A_{21}$ are sets representing the faculties of 21 schools. Then, of course, faculties of all these 21 schools are faculties of IGNOU. That is, all the members of these 21 schools are present in the set U. Here U plays the role of universal set for the sets $A_1, A_2, A_3, \dots, A_{21}$.

A set U is said to be **universal set** if all the sets under study are subsets of U.

Remark 6:

- It may be noted that for different problem-types or different contexts, there will be different universal sets.

- By definition of universal set U , every set X is a subset of U , i.e. $X \subseteq U$. Also, from the fact that $X \subseteq X$ for every X , we get $U \subseteq U$.
- The concept of '*Universal set*', since its inception, has been both controversial and problematic. Specially, it has led to a number of well-known paradoxes including the Russell's paradox, which implies that X , the collection of all sets, is not a well-defined collection, and hence, X is not a set.

However, the concept of *Universal Set* has facilitated representation and solution of problems from most of the domains of our concern including from mathematics itself.

For example,

- If $A = \{1, 2, 3, 5, 7, 9\}$, $B = \{2, 4, 6, 8, 10\}$, $C = \{3, 5, 7\}$, $D = \{8, 9, 10\}$, then $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ can play the role of universal set.
- If in a study, only integers are involved as the elements of the sets, then Z , the set all integers, is the universal set.

Check Your Progress -2

1) Give reasons whether the following statements are true or false:

- If $A = \{2, 9, 7, 7, 5\}$, $B = \{5, 2, 2, 9, 7\}$, then $A = B$

.....

.....

- If $A = \{\alpha, \beta, \gamma\}$, $B = \{\lambda, \mu, \nu\}$, then $A \sim B$

.....

.....

- If $A = \{4, -4, 5, 5\}$ and $B = \{x : \text{either } x^2 = 16 \text{ or } x^2 + x - 20 = 0, x \in Z\}$, then $A = B$

.....

.....

- If $A = \{a, a, b, b, b, c\}$ and $B = \{d, e, e, f, g, h\}$, then $A \sim B$

.....

.....

2) If $A = \{a, b, c, d\}$, then write $P(A)$.

.....

.....

3) If $A = \{1, 3, 5, 7, 9\}$, $B = \{2, 4, 6, 8\}$, $C = \{2, 3, 5, 7, 11, 13, 17, 19, 23\}$, and $D = \{5, 10, 15, 20, 25\}$, then what will be the smallest universal set

.....

.....

1.5 VENN DIAGRAMS

As we know that examples play an important role to understand the concepts of theory/definitions. Similarly a diagram speaks more than the words that we

may use and they also make the ideas simple and easy to understand even for a fresh reader. Euler (1707-1783), a Swiss mathematician was the first who took the step to represent the sets diagrammatically. Then John Venn (1834-1923), a British mathematician, moving a step ahead simplifying the ideas and made it more user friendly. That is why diagrammatical representation of sets is also known as Venn-Euler diagram. But usually they are known as Venn-diagrams. In Venn diagrams, sets are represented by enclosed areas in a plane as described below:

Notations Used in Venn Diagrams

1. Universal Set

Universal set U is represented by the interior of a rectangle as shown in Fig. 1.1

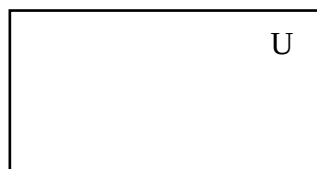
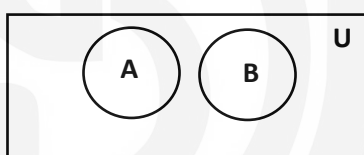


Fig. 1.1: Universal Set

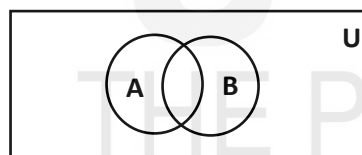
2. Subsets

Subsets of U are described by the interior of closed curves (known as circular discs) within the rectangle, representing the universal set U . Fig. 1.2 shows the case when A and B have no common element, while Fig.1.3 shows the case when A and B have some common elements. Fig. 1.4 shows the case when $A \subseteq B$, and Fig. 1.5 shows the case when $B \subseteq A$.



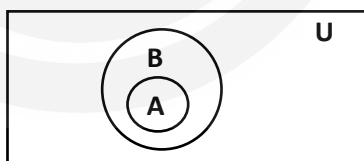
If $A \cap B = \emptyset$

Fig. 1.2: Subset



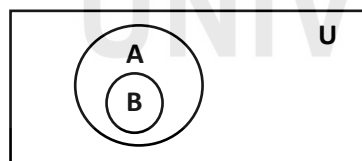
If $A \cap B \neq \emptyset$

Fig. 1.3: Subset



If $A \subseteq B$

Fig. 1.4: Subset



If $B \subseteq A$

Fig. 1.5: Subset

Remark 7:

- (i) In general when nothing is mentioned about the common elements of A and B , presentation of Fig. 1.3 is used.
- (ii) Sizes that we use to present A and B do not matter.

1.6 SET OPERATIONS

In school days, a child first learns counting numbers and then he/she learns how operations of addition, subtraction, multiplication and division are used on two numbers.

1.6.1 Different Set Operations

A similar type of approach is being used here. In sections 1.2 and 1.3 of this unit you have become familiar with the definition and types of sets respectively. In this section, we will learn about some commonly used operations on sets.

Union

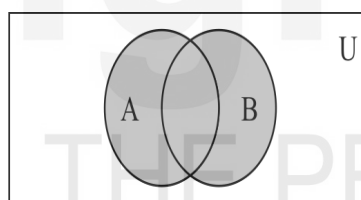
Let A be a set containing the persons getting salary (in Rs) between 10000 and 100000 per month and another set B containing the persons getting salary (in Rs) between 5000 and 20000 per month. For this example, if we are interested in finding those persons who are getting the salary within the range 10000-100000 or 5000-20000 then such persons will be those having salary between 5000 and 100000. The set of such persons is nothing but the union of two sets A and B .

Now, let us formally define the union of two sets.

Let A and B be two sets then union of A and B is denoted by $A \cup B$ and is defined as

$$A \cup B = \{x: \text{either } x \in A \text{ or } x \in B\}.$$

i.e. $A \cup B$ contains all the elements of A as well as of B (see Fig. 1.6).



Shaded region denotes $A \cup B$

Fig. 1.6: Set Operation $A \cup B$

For example,

- (i) If $A = \{2, 3, 5\}$, $B = \{3, 5, 7, 11\}$, then $A \cup B = \{2, 3, 5, 7, 11\}$.
- (ii) If $A = \{a, b, c, d\}$, $B = \{d, e, f\}$, then $A \cup B = \{a, b, c, d, e, f\}$.
- (iii) If $\mathbb{Q}$ = set of all rational numbers and $\mathbb{I}$ = set of all irrational numbers, then $\mathbb{Q} \cup \mathbb{I} = \mathbb{R}$ = set of all real numbers.

That is, if all rational and irrational numbers are mixed then that mixture will be the set of real numbers.

Intersection

Let us again consider the example given above. For this example, if we are interested in finding those persons who are getting the salary within the common range then such persons will be those having salary between 10000 and 20000. The set of such persons is nothing but the intersection of two sets A and B .

Now, let us formally define the intersection of two sets.

Let A and B be two sets then intersection of A and B is denoted by $A \cap B$ and is defined as

$A \cap B = \{x: x \in A \text{ and } x \in B\}$.

i.e. $A \cap B$ contains common elements of A and B (see Fig. 1.7).

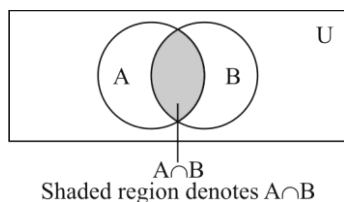


Fig. 1.7 Set Operation $A \cap B$

For example,

(i) If $A = \{a, b, c\}$, $B = \{b, c, d, e\}$, then $A \cap B = \{b, c\}$.

(ii) If $A = \{5, 7, 9\}$, $B = \{10, 11, 18\}$, then $A \cap B = \{ \} = \emptyset =$ empty set, as there is no common element in the two sets.

Note: If $A \cap B = \emptyset$ then we say that two sets A and B are **disjoint**.

Complement of a Set

Suppose we have set of persons of a locality having voting right. Then set of those persons of the locality who do not have voting right is its complement, if the set of all persons of that locality is considered as a universal set.

Now, let us formally define complement of a set.

Let U be the universal set, then complement of a set A (where $A \subseteq U$) is denoted by A^c or $\bar{A}$ or A' and is defined as :

$$A' = \{x \in U: x \notin A\}.$$

That is A' contains those elements of U which are not in A, i.e. A' contains all the elements of U other than A (see Fig. 1.8.).

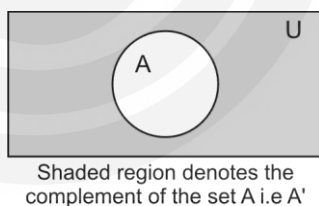


Fig. 1.8: A Complement

For example, if $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ and $A = \{1, 2, 4, 7, 9, 10\}$, then $A' = \{3, 5, 6, 8\}$, i.e. those elements of U which are not in A.

Difference of Two Sets

Let A and B be two sets then difference of A and B is denoted by $A - B$ and is defined as

$A - B = \{x: x \in A \text{ but } x \notin B\}$. See Fig. 1.9

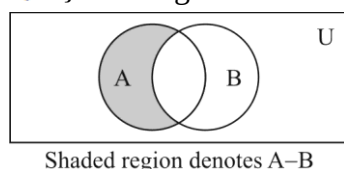


Fig. 1.9: A-B

For example,

(i) If $A = \{1, 2, 3, 4, 5\}$, $B = \{4, 5, 6, 7\}$, then $A - B = \{1, 2, 3\}$

(ii) If $A = \{a, b, c\}$, $B = \{a, b, c, d, e\}$, then $A - B = \{ \}$

Symmetric Difference of Two Sets

Let A and B be two sets, then symmetric difference of A and B is denoted by $A \Delta B$ and is defined as

$A \Delta B = (A - B) \cup (B - A)$. See Fig. 1.10

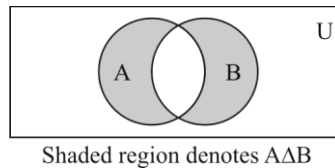


Fig. 1.10: Symmetric Difference to Two Sets

For example, if $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6, 7\}$, then $A - B = \{1, 2\}$,
 $B - A = \{5, 6, 7\}$

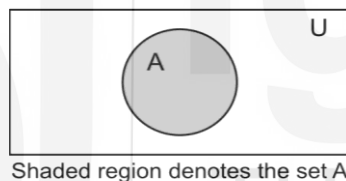
$\therefore A \Delta B = (A - B) \cup (B - A) = \{1, 2\} \cup \{5, 6, 7\} = \{1, 2, 5, 6, 7\}$.

Example 8: With the help of the Venn diagrams, shade the following sets:

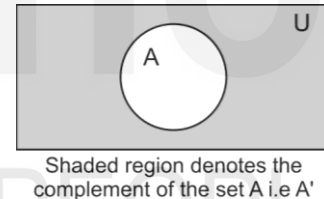
(i) A (ii) A' (iii) $A \cap B$ (iv) $A - B$ (v) $B - A$ (vi) $A \Delta B$ (vii) $A \subseteq B$ (viii) $B \subset A$

Solution:

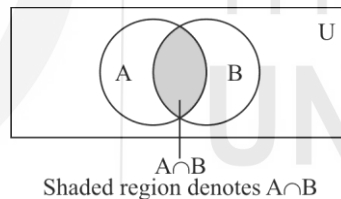
(i)



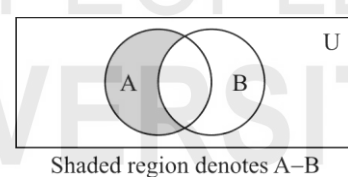
(ii)



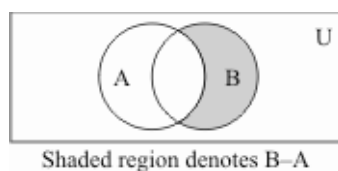
(iii)



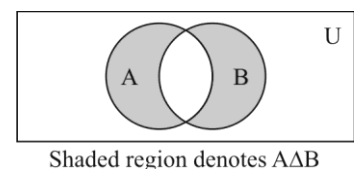
(iv)



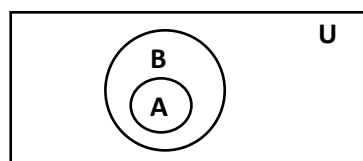
(v)



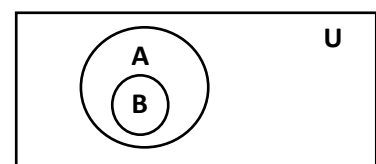
(vi)



(vii)



(viii)



Next, let us we briefly discuss convex set and **Russell's Paradox** .

1.6.2 Convex Set

A convex set in a space (generally 2-dimensional real-space) is a **set** of points such that, given any two points A, B in the **set**, the line AB joining them lies entirely within that **set**.

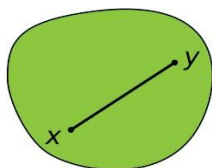


Fig. 1.11: Convex Set Space

Thus, a subset S of a space is called *convex* if and only if for any pair of points p and q in the set S , the line section from p to q is enclosed completely in the set of S , as done in the figure on L.H.S. below.

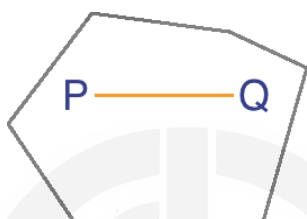


Fig. 1.12: Convex

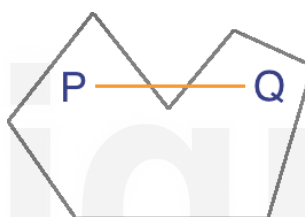


Fig. 1.13: Non-Convex

Algebraically, if x_1 and x_2 are two number in the set S , then all the points on the straight line $x_1 x_2$ – which can be represented as $\lambda x_1 + (1-\lambda) x_2$, for any λ such that $0 \leq \lambda \leq 1$ – also belong to S .

1.6.3 Russell's Paradox

Russell's paradox is a counterexample/ contradiction to naive set theory. In this theory we try to understand working with sets, using natural language rather than formal logic and axioms, which defines a set as any definable collection.

You can construct the Russell's paradox using purely logical statements. This paradox points out the defect in the set theory itself. For example, using the naive set theory, we can construct the set:

$$R = \{x | x \notin x\} ;$$

Here, R is the set of all things x in our mathematical domain (universe) which are not elements of itself. Now the paradox comes here when we try to answer the questions:

Is R itself an element of R ? The answer, of question, is either "yes" or "no". We can examine each case separately.

1. If R is an element of R (i.e. $R \in R$), then R must satisfy the above membership criterion for R ; thus, $R \notin R$. This is of course a contradiction, because by definition R cannot be an element of R .

- Suppose now that R is not an element of R (i.e. $R \notin R$). Then, R satisfies the membership criterion for R , and hence $R \in R$. This is again another contradiction, and this implies that R must be an element of R .

The above two arguments show that: R both can be an element of R and cannot not be an element of R . Which leads to a disastrous and epic contradiction, from which we cannot recover. The same you may infer from the example given below.

A small village has only one (male) barber, who shaves all those, and only those males, who do not shave themselves. Let X be the **collection** of all the villagers shaven by the barber. Then, it is not possible to tell whether barber belongs to X or not. This is so because, if the *barber as a villager* shaves himself, then he cannot be shaven by the barber. Similarly, if *he, as a villager* does not shave himself, then he must be shaven by the barber. Thus, it is not possible to tell whether the barber is in X or not. Hence, X **is not a well-defined collection**, and thus, **not** a set, this is an example of Russell's paradox.

1.6.4 Intervals

Let R be the set of all real numbers. Then a set $I \subseteq R$ is said to be an interval if whenever $a, b \in I$ and $a < x < b$ then $x \in I$

For example, the set of all real numbers satisfying $2 \leq x \leq 3$ is an interval where x can take any real value between 2 and 3 including 2 and 3.

Open Interval

An open interval $I \subseteq R$ with end points a and b ($a < b$) is denoted by (a, b) and is defined by

$$(a, b) = \{x \in R : a < x < b\}$$

i.e. an open interval contains each value between the end points but does not include the end points.

For example, open interval $(2, 5)$ contains each real number lying between 2 and 5 but does not contain 2 and 5.

Closed Interval

A closed interval $I \subseteq R$ with end points a and b ($a < b$) is denoted by $[a, b]$ and is defined by

$$[a, b] = \{x \in R : a \leq x \leq b\}$$

i.e. a closed interval contains each value between and including extreme values.

For example, closed interval $[2, 5]$ contains each real number lying between 2 and 5. It also contains its end points. i.e.

$$x \in [2, 5] \quad \forall x, \quad 2 < x < 5 \text{ and}$$

$$\text{also } 2 \in [2, 5], \quad 5 \in [2, 5]$$

more on intervals you will learn in unit 3 of this course.

Now, you can try the following exercises.

Check Your Progress -3

- 1) If $A = \{3\}$, $B = \{a, b, c\}$, then write $A \cup B$.

2) If $A = \{1, 2, 3\}$, $B = \{3, 4, 5, 6\}$, $C = \{3, 6, 7\}$, then write $A \cup B \cup C$.

3) If $A = \{ \}$, $B = \{1, 4, 7\}$, then write $A \cap B$.

4) If $U = \{x: x \text{ is an English alphabet}\}$ and
 $A = \{x: x \text{ is an vowel of English alphabet}\}$, then write A' .

5) If $W =$ set of whole numbers and $N =$ set of natural numbers then write
 $W - N$.

6) With the help of the Venn diagram justify the following
(i) $A \subseteq A \cup B$ (ii) $B \subseteq A \cup B$ (iii) $A \cap B \subseteq A$ (iv) $A \cap B \subseteq B$
(v) $(A \cap B)' = A' \cup B'$ (De-Morgan's law) (vi) $A \subseteq B \Rightarrow B' \subseteq A'$

7) Tell which one of the following is a convex set, within set of real numbers.

(i) $[4, 5[= \{x: 4 \leq x < 5\}$,

(ii) $[1, 4] \cup [7, 11[= \{x: 1 \leq x \leq 4\} \cup \{x: 7 \leq x < 11\}$

8) What is the Cardinality of the Power set of the set $\{0, 1, 2\}$?

9) Let $U = \{u, v, w, x, y, z\}$,

i) Find the number of subsets of U .

ii) Find the number of proper non-empty subsets of U .

1.7 ORDERED PAIRS AND CARTESIAN PRODUCTS

Sometimes we need to establish relations between two or more sets. For example, a software development company has a set of specialists in different technology domains, or a company gets some projects to develop. Here the company needs to establish a relation between professionals and the project in which they will participate. To solve this type of problem the following concepts are required.

1.7.1 Cartesian product

Very often we deal with several sets at a time, and we need to study their combined action. For instance, combinations of a set of teachers and a set of students. In such a situation we can take a product of these sets to handle them simultaneously. To understand this product let us first consider the following definitions.

Definition: An **ordered pair**, usually denoted by (x,y) , is a pair of elements x and y of some sets. This is ordered in the sense that $(x,y) \neq (y,x)$ whenever $x \neq y$, that is, the order of placing of the element in the pair matters.

Any two ordered pairs (x,y) and (a,b) are equal iff $x = a$ and $y = b$.

For example if, $A = \{a,b,c\}$ and $B = \{x,y,z\}$, then

$A \times B = \{a,b,c\} \times \{x,y,z\} = \{(a,x), (a,y), (a,z), (b,x), (b,y), (b,z), (c,x), (c,y), (c,z)\}$, and

$B \times A = \{x,y,z\} \times \{a,b,c\} = \{(x,a), (x,b), (x,c), (y,a), (y,b), (y,c), (z,a), (z,b), (z,c)\}$.

Now let us think about how $B \times A$ can be represented geometrically?

For instance,

What is the geometric view of $\{2\} \times \mathbb{R}$? This is the line $x = 2$ given in Fig 1.14(a).

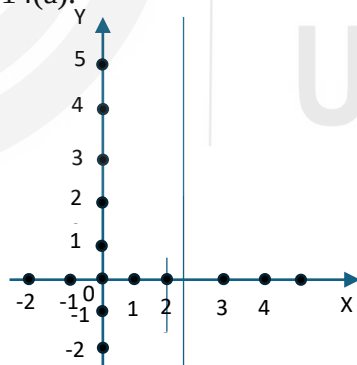


Fig. 1.14:(a) : $\{2\} \times \mathbb{R}$, ie. $x=2$

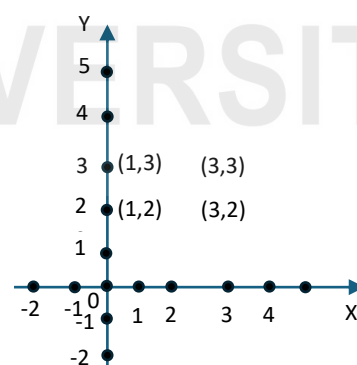


Fig. 1.14: (b): $\{1,3\} \times \{2,3\}$

Now, after seeing geometric representation of $\{2\} \times \mathbb{R}$, can you tell what $\{1,3\} \times \{2,3\} = \{(1,2), (3,2), (1,3), (3,3)\}$ looks like? You will get four points in the first quadrant, as shown in Fig. 1.14(b).

Now, you know that the multiplication of numbers is commutative. Is the Cartesian product of sets also commutative?

For instance, is $\{1\} \times \{2\} = \{2\} \times \{1\}$? No, because $(1,2) \neq (2,1)$. So, $A \times B \neq B \times A$ usually.

We can extend the definition of $A \times B$ to define the Cartesian product of n sets $A_1, A_2, \dots, A_n$ as follows.

$$A_1 \times A_2 \times A_3 \times \dots \times A_n = \{(x_1, x_2, x_3, \dots, x_n)\}:$$

$$x_1 \in A_1, \wedge x_2 \in A_2 \wedge x_3 \in A_3 \wedge \dots, \wedge x_n \in A_n\}.$$

The element $(x_1, x_2, \dots, x_n)$ is called an **n-tuple**. For instance, the 3-tuple $(1, 1, 3) \in \{1\} \times \{1, 2\} \times \{2, 3\}$.

1.8 SOME USEFUL AND IMPORTANT LAWS

Some commonly used laws of sets are listed below.

1. Idempotent Laws

For any set A

$$(i) A \cup A = A \quad (ii) A \cap A = A$$

2. Identity Laws

For any subset A of the universal set U

$$A \cup \emptyset = A, A \cap U = A, \text{ where } \emptyset \text{ is empty set}$$

3. Commutative Laws

For any two sets A and B

$$(i) A \cup B = B \cup A \quad (ii) A \cap B = B \cap A$$

4. Associative Laws

If A, B, C are any three sets then

$$(i) (A \cup B) \cup C = A \cup (B \cup C)$$

$$(ii) (A \cap B) \cap C = A \cap (B \cap C)$$

5. Distributive Laws

If A, B, C are any three sets then

$$(i) A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$(ii) A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

6. Complement:

$$\text{Union: } A \cup A' = U$$

$$\text{Intersection: } A \cap A' = \emptyset$$

7. De-Morgan's Laws

For any two sets A and B the following laws known as De Morgan's laws, hold

$$(i) (A \cup B)' = A' \cap B'$$

$$(ii) (A \cap B)' = A' \cup B'$$

De Morgan's laws can also be expressed as

$$i) A \sim (B \cup C) = (A \sim B) \cap (A \sim C)$$

$$ii) A \sim (B \cap C) = (A \sim B) \cup (A \sim C)$$

Example 9: If $A = \{1, 3, 5\}$, $B = \{3, 5, 7, 9\}$, $C = \{2, 6, 8, 9\}$ are subsets of the universal set $U = \{1, 2, 3, 5, 6, 7, 8, 9\}$ then verify

- (i) De-Morgan's laws and
- (ii) Distributive laws

Solution:

- (i) De-Morgan's laws state that

$$(a) (A \cup B)' = A' \cap B'$$

$$(b) (A \cap B)' = A' \cup B'$$

To verify (a)

$$A \cup B = \{1, 3, 5, 7, 9\}$$

$$\therefore (A \cup B)' = \{2, 6, 8\}$$

$$A' = \{2, 6, 7, 8, 9\}, B' = \{1, 2, 6, 8\}$$

$$\therefore A' \cap B' = \{2, 6, 8\}$$

$$\text{We see that here } (A \cup B)' = A' \cap B'.$$

Hence verified.

To verify (b)

$$A \cap B = \{3, 5\}$$

$$\therefore (A \cap B)' = \{1, 2, 6, 7, 8, 9\} \text{ and}$$

$$A' \cup B' = \{1, 2, 6, 7, 8, 9\}$$

$$\text{We see that here } (A \cap B)' = A' \cup B'.$$

Hence verified.

- (ii) Distributive laws state that

$$(a) A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$(b) A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

To verify (a)

$$B \cap C = \{9\}$$

$$\therefore A \cup (B \cap C) = \{1, 3, 5, 9\}$$

$$A \cup B = \{1, 3, 5, 7, 9\}, A \cup C = \{1, 2, 3, 5, 6, 8, 9\}$$

$$\therefore (A \cup B) \cap (A \cup C) = \{1, 3, 5, 9\}$$

$$\text{We see that here } A \cup (B \cap C) = (A \cup B) \cap (A \cup C).$$

Hence verified.

To verify (b)

$$B \cup C = \{2, 3, 5, 6, 7, 8, 9\}$$

$$\therefore A \cap (B \cup C) = \{3, 5\}$$

$$A \cap B = \{3, 5\}, A \cap C = \{ \}$$

$$\therefore (A \cap B) \cup (A \cap C) = \{3, 5\}$$

$$\text{We see that here } A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Hence verified.

Duality Principle: The 'duality principle' is very convenient for dealing with theorems about sets. Basically if any theorem is given to you, by applying the duality principle you can get another theorem (the dual of the previous one). In any statement involving the union and intersection of sets, we can get from one of the statements to the other by interchanging $\cap$ with $\cup$ and $\emptyset$ with U.

For example, the dual of $A \cup (B \cap C)$ is $A \cap ((B \cup C))$ and the dual of $U \cup \emptyset = U$ is $U \cap \emptyset = \emptyset$. So, for example what is true for $A \cup (B \cap C)$ will be true for $A \cap (B \cup C)$ too. This is why if the first property in each of the pairs below is proved the second one follows immediately.

For any universal set U and subsets A , B and C of U , the following properties hold.

1.9 APPLICATION OF SETS

Venn diagrams are helpful in establishing many important relations between different sets, some of them are mentioned as under which are helpful in solving many practical problems too.

1. $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
2. $n(A \cup B) = n(A - B) + n(A \cap B) + n(B - A)$
3. $n(A - B) = n(A) - n(A \cap B)$
4. $n(B - A) = n(B) - n(A \cap B)$
5. $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$

Let us consider an example based on above formulae.

Example 10: In a group of 500 persons, 400 can speak Hindi and 150 can speak English. Then how many can speak

- (i) both Hindi and English
- (ii) Hindi only
- (iii) English only

Solution: Let A and B denote the set of persons who can speak Hindi and English, respectively. Then in usual notations, we are given

$$n(A \cup B) = 500, n(A) = 400, n(B) = 150$$

- (i) We know that

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$\Rightarrow 500 = 400 + 150 - n(A \cap B) \Rightarrow n(A \cap B) = 550 - 500 = 50$$

$$\therefore \text{number of persons who can speak both Hindi and English} = 50$$
- (ii) We know that

$$n(A - B) = n(A) - n(A \cap B) = 400 - 50 = 350$$

$$\therefore \text{number of persons who can speak Hindi only} = 350$$
- (iii) We know that

$$n(B - A) = n(B) - n(A \cap B) = 150 - 50 = 100$$

$$\therefore \text{number of persons who can speak English only} = 100$$

Check Your Progress -4

- 1) If $X = \{a, b, c\}$ and $Y = \{1, 2, 3\}$, find
 i) $X \times X$, ii) $X \times Y$, and iii) $X \times \emptyset$.

2) Under what conditions on A and B is $A \times B = B \times A$?

3) If the universal set $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$ and $A = \{1, 3, 5, 6, 7\}$,
 $B = \{5, 6, 7, 8\}$, $C = \{1, 5, 7, 8\}$ are subsets of U, then verify
(i) commutative laws and (ii) associative laws

4) Out of the 50 students in a class, 24 play cricket, 15 play hockey, 18 play football, 6 play cricket and hockey, 8 play cricket and football, 5 play hockey and football and 10 students do not play any of the three games. Then how many play (i) all the three games, (ii) hockey but not football and (iii) cricket and football but not hockey.

1.10 SUMMARY

This Unit has provided a foundational overview of **Set Theory**, beginning with methods for representing collections through roster and set-builder notation, then moving to the structural hierarchy of subsets, power sets, and cardinality. Subsequently, visual logic and abstract operations, using Venn Diagrams to demonstrate set interactions such as unions and intersections, are discussed. Also advanced logical concepts such as **Russell's Paradox** and the geometry of **Convex Sets** has been introduced in this unit. The unit concluded with details on Cartesian Products and the application of sets to solve complex real-world problems.

1.11 SOLUTIONS/ANSWERS TO CHECK YOUR PROGRESS

Check Your Progress -1

- 1) (i) It is not a set, because a student may be intelligent according to someone while the same student may not be intelligent according to some other person.
- (ii) It is not a set because a hockey player may be good in someone's view while the same player may not be good in view of some other person.

(iii) It is not a set because an actor may be good in someone's point of view while the same actor may not be good in the view of some other person.

(iv) Set of vowels $\{A, I\}$

2) (i) $x = 2n + 3, n \in W = \{0, 1, 2, 3, 4, \dots\}$

Values are obtained on putting $n = 0, 1, 2, \dots$. In $x = 2n + 3$ e.g. when we put $n = 0$ we get $x = 2(0) + 3 = 3$.

$$\therefore x = 3, 5, 7, 9, 11, \dots$$

And, therefore, $A = \{3, 5, 7, 9, 11, \dots\}$

(ii) $B = \{7^0, 7^1, 7^2, 7^3\} = \{1, 7, 49, 343\}$

(iii) $N = \{1, 2, 3, 4, \dots\}$ and $W = \{0, 1, 2, 3, 4, \dots\}$

$\therefore$ if $x \in N$ and $x \in W$ then the elements which satisfy both

are 1, 2, 3, 4, ... and hence

$$C = \{1, 2, 3, 4, 5, \dots\}.$$

(iv) $W = \{0, 1, 2, 3, \dots\}$ and

$\mathbb{Q}$ = set of rational numbers,

$\therefore$ the elements which are common to both are 0, 1, 2, 3, ... as no other rational number is a whole number and hence

$$D = \{0, 1, 2, 3, \dots\}.$$

3) (i) $\{x \in \mathbb{N} : x = 5n, n \in \mathbb{N}\}$

(ii) $\{x : x = \frac{1}{n}, n \in \mathbb{N}\}$

(iii) $\{x \in \mathbb{N} : x = 2n, n \in \mathbb{N}\}$

4) (i) $X = Y$

(ii) $X \neq Y$ as $15 \in Y$ and $15 \notin X$

(iii) $X = Y$

5) i) False

ii) False

iii) False

(iv) True

(v) True

(vi) False

(vii) True

6) Set of vowels of English alphabet $V = \{a, e, i, o, u\}$.

Two subsets of set V are $V_1 = \{a, e\}$, and $V_2 = \{i, o\}$.

Two supersets of V are $V_3 = \{a, b, c, \dots, z\}$, and $V_4 = \{a, b, c, d, e, i, o, u, \dots, z\}$.

7) Powerset of $A = \{a, e, i, o, u\}$ is :

$\emptyset, \{a\}, \{e\}, \{i\}, \{o\}, \{u\}, \{a, e\}, \{a, i\}, \{a, o\}, \{a, u\}, \{e, i\}, \{e, o\}, \{e, u\}, \{i, o\}, \{i, u\}, \{o, u\}, \{a, e, i\}, \{a, e, o\}, \{a, e, u\}, \{a, i, u\}, \{a, i, o\}, \{a, o, u\}, \{e, i, o\}, \{e, i, u\}, \{i, o, u\}, \{e, o, u\}, \{a, e, i, u\}, \{a, e, i, o\}, \{e, i, o, u\}, \{a, e, o, u\}, \{a, i, o, u\}, \{a, e, i, o, u\}$.

8) For empty set $A = \{\}$ or $\emptyset$, $P(A) = 1$

9) If $A \subset B$, then $P(A) \subset P(B)$ because every subset of A is a subset of B .

10) If $P(A) = P(B)$ then $A \in P(A) = P(B) = A \subset B$. Similarly, $B \subset A$. Therefore $A = B$.

Check Your Progress -2

1) (i) $A = \{2, 9, 7, 5\}$, $B = \{5, 2, 9, 7\}$ as repetitions in a set are not allowed.

We see that all the elements of A are in B and all the elements of B are in A .

$\therefore A = B$. Hence the statement is true.

(ii) Here, $n(A) = 3$, $n(B) = 3$ and so the statement $A \sim B$ is true.

(iii) $A = \{4, -4, 5\}$, $B = \{x: x^2 = 16, x^2 + x - 20 = 0, x \in \mathbb{Z}\}$
 $= \{x: x = \pm 4, (x+5)(x-4) = 0, x \in \mathbb{Z}\}$
 $= \{x: x = \pm 4, x = -5, 4, x \in \mathbb{Z}\} = \{4\}$

Here, $-4 \in A$ but $-4 \notin B$

$\therefore A \neq B$, hence the statement is false.

(iv) Here, $A = \{a, b, c\}$, $B = \{d, e, f, g, h\}$. Thus, $n(A) = 3$, $n(B) = 5$.

$\therefore A \sim B$ is false because $n(A) \neq n(B)$.

2) $P(A) = \{\varnothing, \{a\}, \{b\}, \{c\}, \{d\}, \{a, b\}, \{a, c\}, \{a, d\}, \{b, c\}, \{b, d\}, \{c, d\}, \{a, b, c\}, \{a, b, d\}, \{a, c, d\}, \{b, c, d\}, \{a, b, c, d\}\}$

Notice that it has $2^4 = 16$ elements.

3) Smallest universal set is

$\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 13, 15, 17, 19, 20, 23, 25\}$.

Check Your Progress -3

1) $A \cup B = \{3, a, b, c\}$.

2) $A \cup B \cup C = \{1, 2, 3, 4, 5, 6, 7\}$.

3) $A \cap B = \{\} = \varnothing$.

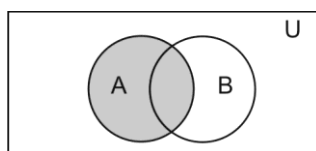
4) $A' = \{x: x \text{ is a consonant of English alphabet}\}$.

5) We know that

$W = \{0, 1, 2, 3, 4, \dots\}$ and $N = \{1, 2, 3, 4, \dots\}$

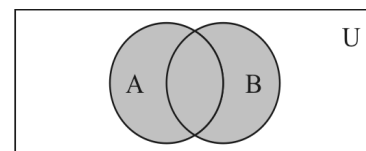
$\therefore W - N = \{0, 1, 2, 3, 4, \dots\} - \{1, 2, 3, 4, \dots\} = \{0\}$

6) (i)



Shaded region denotes A

L.H.S.

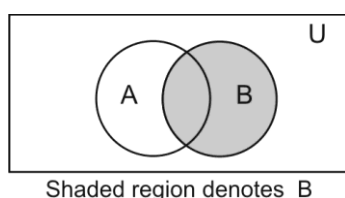


Shaded region denotes $A \cup B$

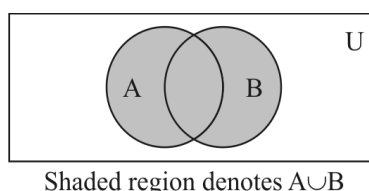
R.H.S.

Two Venn diagrams justify the relationship $A \subseteq A \cup B$.

(ii)



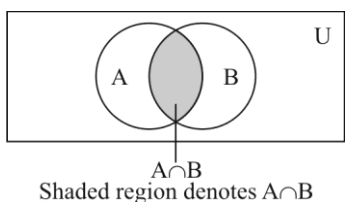
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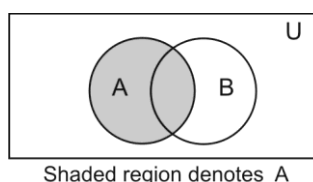
R.H.S.

Two Venn diagrams justify the relationship $B \subseteq A \cup B$.

(iii)



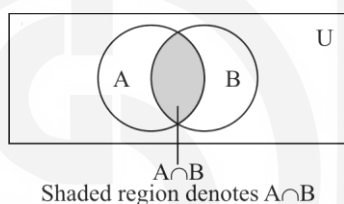
L.H.S.



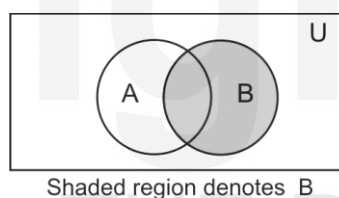
R.H.S.

Two Venn diagrams justify the relationship $A \cap B \subseteq A$.

(iv)



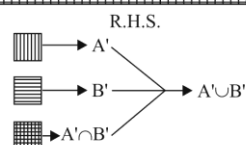
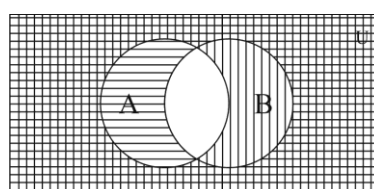
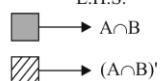
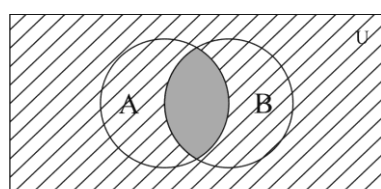
L.H.S.



R.H.S.

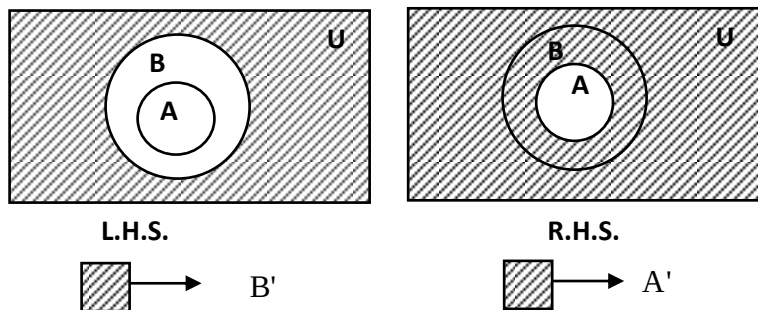
Two Venn diagrams justify the relationship $A \cap B \subseteq B$.

(v)



Two Venn diagrams justify the relationship $(A \cap B)' = A' \cup B'$.

(vi)



Two Venn diagrams justify the relationship $A \subseteq B \Rightarrow B' \subseteq A'$.

7)

- i) it is convex
- ii) it is not convex

8) $2^3 = 8$

9)

- i) There are $2^6 = 64$ subsets
- ii) The answer is 62: there are $2^6 = 64$ subsets in U altogether. We exclude two: U itself (because it is not proper subset) and the empty set (because it is not non-empty).

Check Your Progress -4

1) i) $X \times X = \{(a,a), (a,b), (a,c), (b,b), (b,c), (c,c)\}$.

ii) $X \times Y = \{(a,1), (b,1), (c,1), (a,2), (b,2), (c,2), (a,3), (b,3), (c,3)\}$.

ii) $X \times X \setminus \emptyset = \emptyset$.

2) $A \times B = B \times A$ iff $A = B$ or one of A, B is $\emptyset$.

3) (i) Commutative laws state that

(a) $A \cup B = B \cup A$

(b) $A \cap B = B \cap A$

To verify (a)

$$A \cup B = \{1, 3, 5, 6, 7\} \cup \{5, 6, 7, 8\} = \{1, 3, 5, 6, 7, 8\}$$

$$\therefore B \cup A = \{5, 6, 7, 8\} \cup \{1, 3, 5, 6, 7\} = \{1, 3, 5, 6, 7, 8\}$$

$\therefore$ we see that here $A \cup B = B \cup A$.

Hence verified.

To verify (b)

$$A \cap B = \{1, 3, 5, 6, 7\} \cap \{5, 6, 7, 8\} = \{5, 6, 7\}$$

$$\text{and } B \cap A = \{5, 6, 7, 8\} \cap \{1, 3, 5, 6, 7\} = \{5, 6, 7\}$$

We see that here $A \cap B = B \cap A$

Hence verified.

(ii) Associative laws state that

$$(a) (A \cup B) \cup C = A \cup (B \cup C)$$

$$(b) (A \cap B) \cap C = A \cap (B \cap C)$$

To verify (a)

$$A \cup B = \{1, 3, 5, 6, 7, 8\}$$

$$\therefore (A \cup B) \cup C = \{1, 3, 5, 6, 7, 8\}$$

$$B \cup C = \{5, 6, 7, 8\} \cup \{1, 5, 7, 8\} = \{1, 5, 6, 7, 8\}$$

$$\therefore A \cup (B \cup C) = \{1, 3, 5, 6, 7\} \cup \{1, 5, 6, 7, 8\} = \{1, 3, 5, 6, 7, 8\}$$

We see that here $(A \cup B) \cup C = A \cup (B \cup C)$.

Hence verified.

To verify (b)

$$A \cap B = \{5, 6, 7\}$$

$$\therefore (A \cap B) \cap C = \{5, 6, 7\} \cap \{1, 5, 7, 8\} = \{5, 7\}$$

$$B \cap C = \{5, 6, 7, 8\} \cap \{1, 5, 7, 8\} = \{5, 7, 8\}$$

$$\therefore A \cap (B \cap C) = \{1, 3, 5, 6, 7\} \cap \{5, 7, 8\} = \{5, 7\}$$

We see that here $(A \cap B) \cap C = A \cap (B \cap C)$.

Hence verified.

- 4) Let C, H, F, denote the set of students who play cricket, hockey and football respectively. Then in usual notations, we are given.
 $n(C) = 24$, $n(H) = 15$, $n(F) = 18$, $n(C \cap H) = 6$, $n(C \cap F) = 8$,
 $n(H \cap F) = 5$, $n(C' \cap H' \cap F') = 10$

- (i) Before finding the required number of students, we are to first obtain the number of students who play at least one of the three games which is given as
 $= n(C \cup H \cup F) = 50 - n(C' \cap H' \cap F') = 50 - 10 = 40$

Now, we know that

$$n(C \cup H \cup F) = n(C) + n(H) + n(F) - n(C \cap H) - n(C \cap F) - n(H \cap F) + n(C \cap H \cap F)$$

$$\Rightarrow 40 = 24 + 15 + 18 - 6 - 8 - 5 + n(C \cap H \cap F)$$

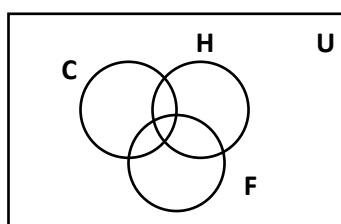
$$\Rightarrow 40 = 57 - 19 + n(C \cap H \cap F) = 38 + n(C \cap H \cap F)$$

$$\Rightarrow n(C \cap H \cap F) = 40 - 38 = 2$$

$\therefore$ number of students who play all the three games = 2

- (ii) We know that

$$\begin{aligned} n(H - F) &= n(H) - n(H \cap F) \\ &= 15 - 5 = 10 \end{aligned}$$



$\therefore$ number of students who play hockey but not football = 10

$$(iii) n((C \cap F) - H) = n(C \cap F) - n((C \cap F) \cap H) = 8 - 2 = 6$$

$$\left[\because n(A - B) = n(A) - n(A \cap B) \right]$$

$$[Here A = C \cap F, B = H]$$

$\therefore$ number of students who play cricket and football but not hockey = 6

1.12 REFERENCES / FURTHER READINGS

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